Lecture 17

Dimensions and weight multiplicities

Objectives. *The Weyl character formula writes* $\operatorname{ch} L(\lambda)$ *as a quotient of alternants. We turn it into three calculations: the dimension of* $L(\lambda)$*, each weight multiplicity, and the decomposition of a given finite-dimensional module.*

By the end of this lecture you will be able to:

- Derive the Weyl dimension formula by a limit along a regular direction.
- Derive Kostant's multiplicity formula and find its contributing Weyl group terms.
- Decompose a finite-dimensional module by subtracting irreducible characters.
- Carry out these calculations for $\mathfrak{sl}_3$.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra with the fixed data $\mathfrak{h}$, R_+, simple roots $\alpha_1, \dots, \alpha_r$, lattices $Q \subseteq P$, dominant weights P_+ and ρ. Except for the cited Verma modules, modules are finite-dimensional. Write Q_+ for the set of nonnegative integer combinations of simple roots. The partial order $\leq$ on P is

$$\mu \leq \nu \quad \Longleftrightarrow \quad \nu - \mu \in Q_+.$$

17.1 Why evaluation needs a limit

For $\lambda \in P_+$ and $\mu \in P$ write

$$m_\lambda(\mu) = \dim L(\lambda)_\mu.$$

Lecture 16 proved the Weyl character formula. For $\nu \in P$, it uses the alternant

$$A_\nu = \sum_{w \in W} \det(w)\, e^{w\nu}.$$

Each alternant is a finite sum, so the character formula is an identity in the group ring $\mathbb{Z}[P]$:

$$\operatorname{ch} L(\lambda) \cdot A_\rho = A_{\lambda+\rho}. \tag{17.1}$$

Lecture 15 proved the denominator identity

$$A_\rho = e^\rho \prod_{\alpha \in R_+} \left(1 - e^{-\alpha}\right). \tag{17.2}$$

Fix the W-invariant positive definite form $(\ ,\ )$ on the real span of the roots (Lecture 8). The weight lattice lies in this span. For $\mu \in P$ and $\alpha \in R$ the coroot pairing is

$$\langle \mu, \alpha^\vee \rangle = \mu(h_\alpha) = \frac{2(\mu, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

The form identifies the coroot h_α with $2\alpha/(\alpha,\alpha)$ in the real root span. Recall that $\langle \rho, \alpha_i^\vee \rangle = 1$ for every simple root, so $\rho = \omega_1 + \cdots + \omega_r$ (Lecture 15).

The dimension of a module is the sum of the coefficients of its character. This sum is the image under the ring homomorphism

$$\mathbb{Z}[P] \longrightarrow \mathbb{Z}, \qquad e^\mu \longmapsto 1.$$

Applied to the character formula, this homomorphism gives no information. Since $\mathfrak{g} \neq 0$, the root set R is nonempty, so W contains a reflection s. The map $w \mapsto sw$ is a bijection of W that reverses det. Hence

$$\sum_{w \in W} \det(w) = 0.$$

Every alternant maps to 0, and (17.1) becomes $0 = 0$. We introduce a parameter t and compare leading terms. At $t = 0$, the new evaluation will send every e^μ to 1.

17.2 The Weyl dimension formula

Definition 17.1 (Specialisation along a weight)**.** For $\nu \in P$, let $\Phi_\nu \colon \mathbb{Z}[P] \to \mathbb{C}[[t]]$ be the additive map with

$$\Phi_\nu(e^\mu) = \exp\bigl(t(\mu,\nu)\bigr) = \sum_{k \geq 0} \frac{(\mu,\nu)^k}{k!}\, t^k.$$

The weight ν is *regular* if $(\alpha,\nu) \neq 0$ for every $\alpha \in R$.

The binomial theorem gives $\exp(at)\exp(bt) = \exp((a+b)t)$ in $\mathbb{C}[[t]]$, so Φ_ν is a ring homomorphism. Write $O(t^k)$ for an element of $t^k\mathbb{C}[[t]]$. The constant term of $\Phi_\nu(f)$ is the coefficient sum of f. Hence, for every finite-dimensional module V,

$$\Phi_\nu(\operatorname{ch} V) = \dim V + O(t). \tag{17.3}$$

Lemma 17.2 (Symmetry · *P*)**.** *For $\mu, \nu \in P$, we have $\Phi_\nu(A_\mu) = \Phi_\mu(A_\nu)$.*

Proof. Use invariance and symmetry of the form. Then reindex by w^{-1}, whose determinant is $\det(w)$:

$$\begin{aligned}
\Phi_\nu(A_\mu) &= \sum_{w \in W} \det(w) \exp\bigl(t(w\mu,\nu)\bigr) \\
&= \sum_{w \in W} \det(w) \exp\bigl(t(\mu, w^{-1}\nu)\bigr) \\
&= \sum_{w \in W} \det(w^{-1}) \exp\bigl(t(w^{-1}\nu,\mu)\bigr) \\
&= \sum_{w \in W} \det(w) \exp\bigl(t(w\nu,\mu)\bigr) \\
&= \Phi_\mu(A_\nu). \qquad \blacksquare
\end{aligned}$$

Lemma 17.3 (Leading term of the denominator · *P*)**.** *Let $N = |R_+|$. For $\nu \in P$,*

$$\Phi_\nu(A_\rho) = t^N \prod_{\alpha \in R_+} (\alpha,\nu) + O(t^{N+1}).$$

Proof. Apply Φ_ν to the product (17.2). For a number c,

$$1 - \exp(-ct) = ct - \frac{c^2t^2}{2} + \cdots = t\big(c + O(t)\big).$$

Therefore

$$\begin{aligned}\Phi_\nu(A_\rho) &= \exp\big(t(\rho,\nu)\big) \prod_{\alpha\in R_+} \Big(1 - \exp\big(-t(\alpha,\nu)\big)\Big) \\ &= \big(1 + O(t)\big)\, t^N \prod_{\alpha\in R_+} \big((\alpha,\nu) + O(t)\big).\end{aligned}$$

Expanding the product gives the claim. ∎

Since $(-\alpha,\nu) = -(\alpha,\nu)$, the leading coefficient is nonzero exactly when ν is regular. The weight ρ is regular. To see this, first compute, for each simple root α_i,

$$(\rho,\alpha_i) = \frac{1}{2}(\alpha_i,\alpha_i)\,\langle\rho,\alpha_i^\vee\rangle = \frac{1}{2}(\alpha_i,\alpha_i) > 0. \tag{17.4}$$

A positive root has the form $\alpha = \sum_i c_i\alpha_i$ with integers $c_i \geq 0$, not all zero. Hence $(\rho,\alpha) > 0$ for every $\alpha \in R_+$.

Theorem 17.4 (Weyl dimension formula · *P*)**.** *For $\lambda \in P_+$,*

$$\dim L(\lambda) = \prod_{\alpha\in R_+} \frac{(\lambda+\rho,\alpha)}{(\rho,\alpha)} = \prod_{\alpha\in R_+} \frac{\langle\lambda+\rho,\alpha^\vee\rangle}{\langle\rho,\alpha^\vee\rangle}.$$

Proof. Apply the ring homomorphism Φ_ρ to (17.1):

$$\Phi_\rho\big(\operatorname{ch} L(\lambda)\big)\,\Phi_\rho(A_\rho) = \Phi_\rho(A_{\lambda+\rho}).$$

By (17.3) and Lemma 17.3 with $\nu = \rho$, the left side equals

$$\begin{aligned}&\big(\dim L(\lambda) + O(t)\big)\Big(t^N \prod_{\alpha\in R_+} (\alpha,\rho) + O(t^{N+1})\Big) \\ &= t^N \dim L(\lambda) \prod_{\alpha\in R_+} (\alpha,\rho) + O(t^{N+1}).\end{aligned}$$

By Lemma 17.2, and then Lemma 17.3 with $\nu = \lambda + \rho$, the right side equals

$$\Phi_\rho(A_{\lambda+\rho}) = \Phi_{\lambda+\rho}(A_\rho) = t^N \prod_{\alpha\in R_+} (\alpha,\lambda+\rho) + O(t^{N+1}).$$

Compare the coefficients of t^N. The numbers (α,ρ) are positive, as shown after (17.4), so we may divide by their product. This gives the first formula. Multiplying the numerator and denominator of each factor by $2/(\alpha,\alpha)$ gives the second. ∎

The argument is also a limit in the analytic sense. For $f \in \mathbb{Z}[P]$, the series $\Phi_\nu(f)$ is the Taylor series at 0 of a finite sum of exponentials, an entire function of t. Since ρ is regular, the denominator $\Phi_\rho(A_\rho)$ vanishes at $t = 0$ to order exactly N. The proof shows that the numerator $\Phi_\rho(A_{\lambda+\rho})$ is t^N times a power series. Hence

$$\dim L(\lambda) = \lim_{t\to 0} \frac{\Phi_\rho(A_{\lambda+\rho})(t)}{\Phi_\rho(A_\rho)(t)}.$$

Along a nonregular ν, some factor $1-\exp\bigl(-t(\alpha,\nu)\bigr)$ in the proof of Lemma 17.3 is 0. Hence $\Phi_\nu(A_\rho)=0$ identically (Exercise 17.5).

Each factor is a ratio of two values of the form at the same root α. Hence rescaling the form leaves the dimension formula unchanged.

Now write $\lambda=\sum_i m_i\omega_i$. Suppose every simple component of $\mathfrak{g}$ is simply laced, that is, its roots have the same length (types A, D, E). A positive root $\alpha=\sum_i c_i\alpha_i$ involves only simple roots of its own component, so $(\alpha,\alpha)=(\alpha_i,\alpha_i)$ whenever $c_i\neq 0$. Hence $\alpha^\vee=2\alpha/(\alpha,\alpha)=\sum_i c_i\alpha_i^\vee$. Since $\langle\lambda+\rho,\alpha_i^\vee\rangle=m_i+1$, we get

$$\alpha=\sum_i c_i\alpha_i \quad\Longrightarrow\quad \langle\lambda+\rho,\alpha^\vee\rangle=\sum_i c_i(m_i+1). \tag{17.5}$$

When a component has two root lengths, this shortcut can fail. Exercise 17.2 shows the resulting error in type C_2.

17.3 Dimensions in $\mathfrak{sl}_3$

Let $\mathfrak{g}=\mathfrak{sl}_3$ with the diagonal Cartan subalgebra. Let ε_i read the ith diagonal entry, so $\varepsilon_1+\varepsilon_2+\varepsilon_3=0$ on $\mathfrak{h}$. The A_2 data are

$$\alpha_1=\varepsilon_1-\varepsilon_2,\quad \alpha_2=\varepsilon_2-\varepsilon_3,\qquad R_+=\{\alpha_1,\alpha_2,\alpha_1+\alpha_2\},$$
$$\omega_1=\varepsilon_1,\qquad \omega_2=\varepsilon_1+\varepsilon_2=-\varepsilon_3,\qquad \rho=\alpha_1+\alpha_2=\omega_1+\omega_2.$$

All roots have the same length. In fundamental-weight coordinates, $\alpha_1=2\omega_1-\omega_2$ and $\alpha_2=-\omega_1+2\omega_2$.

Example 17.5 (The $\mathfrak{sl}_3$ dimension formula)**.** Let $\lambda=p\omega_1+q\omega_2$ with integers $p,q\geq 0$. Then $\lambda+\rho=(p+1)\omega_1+(q+1)\omega_2$. By (17.5),

α	α_1	α_2	$\alpha_1+\alpha_2$
$\langle\lambda+\rho,\alpha^\vee\rangle$	$p+1$	$q+1$	$p+q+2$
$\langle\rho,\alpha^\vee\rangle$	1	1	2

Theorem 17.4 gives

$$\dim L(p\omega_1+q\omega_2)=\frac{(p+1)(q+1)(p+q+2)}{2}.$$

Some small values are

(p,q)	$(0,0)$	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(3,0)$	$(2,2)$
$\dim L(p\omega_1+q\omega_2)$	1	3	3	6	8	10	27

The table passes three checks. First, $L(\omega_1)=\mathbb{C}^3$ (Lecture 13), and the table gives $\dim L(\omega_1)=3$. Next, the adjoint module $\mathfrak{sl}_3$ is irreducible with highest root $\varepsilon_1-\varepsilon_3=\omega_1+\omega_2$. The table gives $\dim L(\omega_1+\omega_2)=8=\dim\mathfrak{sl}_3$. Finally, the formula is symmetric in p and q. This matches $L(\lambda)^*\cong L(-w_0\lambda)$ (Lecture 14), since $-w_0$ exchanges ω_1 and ω_2 for $\mathfrak{sl}_3$.

17.4 Kostant's multiplicity formula

The *Kostant partition function* $p(\gamma)$ is the number of families $(k_\alpha)_{\alpha\in R_+}$ of nonnegative integers with $\sum_\alpha k_\alpha\alpha=\gamma$. Lecture 15 placed Verma characters in a commutative ring $\mathcal{E}$ of formal

sums $\sum c_\mu e^\mu$ supported in finite unions of sets $\nu - Q_+$. This ring contains $\mathbb{Z}[P]$ as a subring, and $\operatorname{ch} M(\lambda) = e^\lambda K$, where

$$K = \sum_{\gamma \in Q_+} p(\gamma)\, e^{-\gamma}.$$

Lemma 17.6 (Inverse of the denominator · *P*)**.** *In $\mathcal{E}$,*

$$\prod_{\alpha \in R_+} \left(1 - e^{-\alpha}\right) \cdot K = 1.$$

Proof. For $\alpha \in R_+$ put

$$G_\alpha = \sum_{k \geq 0} e^{-k\alpha}.$$

In $(1 - e^{-\alpha})G_\alpha$, the constant term is 1, and the coefficient of $e^{-k\alpha}$ is $1 - 1 = 0$ for $k \geq 1$. Hence

$$\left(1 - e^{-\alpha}\right)G_\alpha = 1.$$

The coefficient of $e^{-\gamma}$ in $\prod_{\alpha \in R_+} G_\alpha$ counts the families (k_α) with $\sum k_\alpha \alpha = \gamma$. Thus $\prod_\alpha G_\alpha = K$. Multiply the N identities above, using commutativity and associativity in $\mathcal{E}$. ∎

Theorem 17.7 (Kostant's multiplicity formula · *P*)**.** *For $\lambda \in P_+$ and $\mu \in P$,*

$$m_\lambda(\mu) = \sum_{w \in W} \det(w)\, p\big(w(\lambda + \rho) - (\mu + \rho)\big),$$

where $p(\gamma) = 0$ for $\gamma \notin Q_+$.

Proof. By (17.2), identity (17.1) reads

$$\operatorname{ch} L(\lambda) \cdot e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}) = A_{\lambda+\rho}.$$

Multiply both sides by $e^{-\rho}K$ and use Lemma 17.6:

$$\operatorname{ch} L(\lambda) = e^{-\rho} A_{\lambda+\rho} K = \sum_{w \in W} \det(w) \sum_{\gamma \in Q_+} p(\gamma)\, e^{w(\lambda+\rho)-\rho-\gamma}.$$

Now extract the coefficient of e^μ. For fixed w, the exponent equals μ for exactly one γ, namely

$$\gamma = w(\lambda + \rho) - (\mu + \rho).$$

That term contributes $\det(w)\, p(\gamma)$, and it contributes zero if $\gamma \notin Q_+$. Summing over the $|W|$ elements gives the formula. ∎

The formula is a finite sum of $|W|$ integers. Its terms are controlled by the support of p.

(a) $p(\gamma) > 0$ exactly when $\gamma \in Q_+$. If $\gamma = \sum_i c_i \alpha_i \in Q_+$, then $k_{\alpha_i} = c_i$, with all other $k_\alpha = 0$, gives a family. Conversely, every positive root lies in Q_+, so every sum $\sum_\alpha k_\alpha \alpha$ does too.

(b) The term for w survives only if $\mu + \rho \leq w(\lambda + \rho)$. In practice, list the orbit of $\lambda + \rho$ and discard the points x for which $\mu + \rho \leq x$ fails.

(c) The term for $w = 1$ is $p(\lambda - \mu) = \dim M(\lambda)_\mu$ (Lecture 15). The other terms correct this weight multiplicity of the Verma module.

For A_2 the partition function is explicit. A family for $a\alpha_1 + b\alpha_2$, with $a, b \geq 0$, is fixed by the number k of copies of $\alpha_1 + \alpha_2$. Here $0 \leq k \leq \min(a, b)$, and the remaining summands are $(a-k)\alpha_1 + (b-k)\alpha_2$. Hence

$$p(a\alpha_1 + b\alpha_2) = \min(a, b) + 1 \qquad (a, b \in \mathbb{Z}_{\geq 0}). \tag{17.6}$$

Write (x, y) for $x\omega_1 + y\omega_2$. The simple reflections act by

$$s_1(x, y) = (-x, x + y), \qquad s_2(x, y) = (x + y, -y).$$

Composing them gives the orbit of (x, y):

w	1	s_1	s_2	s_1s_2	s_2s_1	$s_1s_2s_1$
$w(x, y)$	(x, y)	$(-x, x + y)$	$(x + y, -y)$	$(-x - y, x)$	$(y, -x - y)$	$(-y, -x)$
$\det(w)$	$+$	$-$	$-$	$+$	$+$	$-$

The conversion to root coordinates is

$$(x, y) = \frac{2x + y}{3}\,\alpha_1 + \frac{x + 2y}{3}\,\alpha_2.$$

Example 17.8 (The adjoint zero weight)**.** Take $\lambda = \omega_1 + \omega_2$ and $\mu = 0$. Then $\lambda + \rho = (2, 2)$ and $\mu + \rho = (1, 1)$. The arguments $w(2, 2) - (1, 1)$ are

w	1	s_1	s_2	s_1s_2	s_2s_1	$s_1s_2s_1$
ω-coordinates	$(1, 1)$	$(-3, 3)$	$(3, -3)$	$(-5, 1)$	$(1, -5)$	$(-3, -3)$
root coordinates	$(1, 1)$	$(-1, 1)$	$(1, -1)$	$(-3, -1)$	$(-1, -3)$	$(-3, -3)$
p	2	0	0	0	0	0

By (17.6), $p(\alpha_1 + \alpha_2) = 2$. Hence

$$m_{\omega_1+\omega_2}(0) = 2.$$

The zero weight space is the diagonal Cartan subalgebra, of dimension 2. The six roots have multiplicity one, giving $6 + 2 = 8 = \dim \mathfrak{sl}_3$.

Here only $w = 1$ contributes. The next example needs a cancellation. Figure 17.1 draws the support test for it. Only the orbit points above $\mu + \rho$ contribute.

Example 17.9 (A cancelling term)**.** Take $\lambda = 3\omega_1$ and $\mu = 0$. Then $\lambda + \rho = (4, 1)$ and $\mu + \rho = (1, 1)$. The surviving terms are

$$\begin{aligned} w = 1: &\quad (4, 1) - (1, 1) = (3, 0) = 2\alpha_1 + \alpha_2, \quad p = 2, \\ w = s_2: &\quad (5, -1) - (1, 1) = (4, -2) = 2\alpha_1, \qquad p = 1. \end{aligned}$$

The other four arguments are $(-2, 1)$, $(-3, 0)$, $(-2, -4)$ and $(-3, -4)$ in root coordinates. They are not in Q_+. Therefore

$$m_{3\omega_1}(0) = 2 - 1 = 1.$$

The general zero-weight calculation for $L(n\rho)$ is in Exercise 17.6(iv).

Figure 17.2 shows the modules of these two examples with all their weight multiplicities. Kostant's formula gives the values at the dominant weights. Weyl invariance (Lecture 15) then gives the rest.

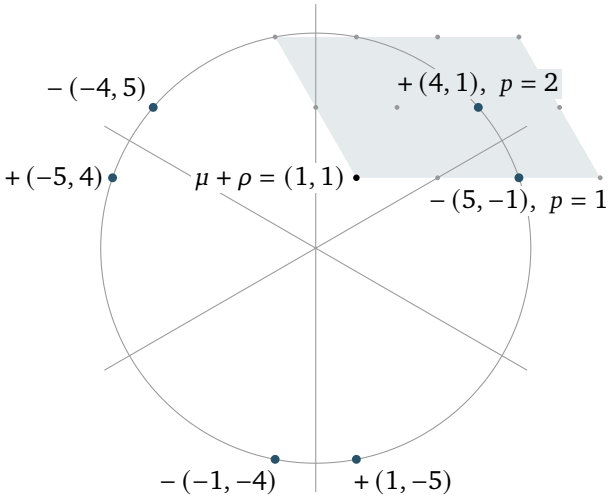


Figure 17.1: The orbit of $\lambda + \rho = (4, 1)$ for $\lambda = 3\omega_1$ on its circle of squared norm 14 for the form with $(\alpha, \alpha) = 2$, with the signs $\det(w)$. The grey dots in the shaded region are points of $(1, 1) + Q_+$, and only two orbit points lie among them, so $m_{3\omega_1}(0) = 2 - 1$.

17.5 Decomposing a module by character subtraction

A difference of characters is an element of $\mathbb{Z}[P]^W$ (Lecture 15). The next result shows how to subtract so that the result is again the character of a module.

Characters are W-invariant, and every orbit in P meets P_+ (Lecture 9). A nonzero module therefore has a dominant weight. Its finite weight set has a maximal dominant weight. It suffices to record multiplicities of dominant weights.

Proposition 17.10 (Character subtraction · *P*)**.** *Let V be a nonzero finite-dimensional $\mathfrak{g}$-module. Write $V \cong \bigoplus_{\nu \in P_+} L(\nu)^{n_\nu}$ (Lecture 5, Lecture 13). Let λ be maximal for $\leq$ among the dominant weights of V. Then $n_\lambda = \dim V_\lambda > 0$, and*

$$\operatorname{ch} V - n_\lambda \operatorname{ch} L(\lambda) = \operatorname{ch} V', \qquad V \cong V' \oplus L(\lambda)^{n_\lambda}.$$

Repeating this step decomposes V. In particular, $\operatorname{ch} V$ determines V up to isomorphism.

Proof. Let $S = \{\nu : n_\nu > 0\}$. Each $\nu \in S$ is a dominant weight of V. The weights of $L(\nu)$ lie in $\nu - Q_+$, and $m_\nu(\nu) = 1$ (Lecture 12).

The weight λ is a weight of some $L(\nu)$ with $\nu \in S$, so $\lambda \leq \nu$. Maximality of λ forces $\nu = \lambda$. Thus $\lambda \in S$.

Next,

$$\dim V_\lambda = \sum_{\nu \in S} n_\nu \, m_\nu(\lambda).$$

A nonzero term needs $\lambda \leq \nu$ with $\nu \in S$. Maximality gives $\nu = \lambda$, so the sum is $n_\lambda m_\lambda(\lambda) = n_\lambda$.

Let $V' = \bigoplus_{\nu \neq \lambda} L(\nu)^{n_\nu}$. Additivity of characters gives the displayed identity.

The weight λ and the number $n_\lambda = \dim V_\lambda$ were read from $\operatorname{ch} V$ alone. Hence $\operatorname{ch} V$ determines $\operatorname{ch} V'$, and $\dim V' < \dim V$ since $n_\lambda > 0$. Induction on $\dim V$, starting from $V = 0$, proves the final assertion. ∎

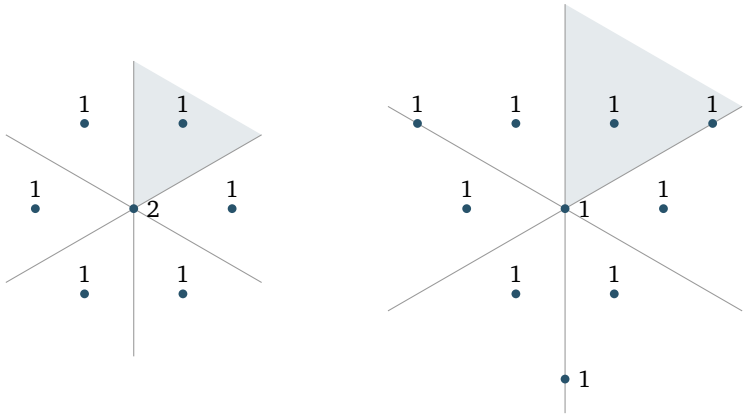


Figure 17.2: Weight multiplicities of the adjoint module $L(\omega_1+\omega_2)$ (left) and of $L(3\omega_1)$ (right), printed at each weight. The dominant chamber is shaded. Weyl invariance determines all multiplicities from those at dominant weights, including points on the walls.

Example 17.11 ($\mathbb{C}^3 \otimes (\mathbb{C}^3)^*$ for $\mathfrak{sl}_3$)**.** The weights of $L(\omega_1)=\mathbb{C}^3$ are $\varepsilon_1,\varepsilon_2,\varepsilon_3$. The dual $L(\omega_2)\cong L(\omega_1)^*$ has weights $-\varepsilon_j$. By multiplicativity (Lecture 15),

$$\operatorname{ch}\bigl(L(\omega_1)\otimes L(\omega_2)\bigr)=\sum_{i,j=1}^{3} e^{\varepsilon_i-\varepsilon_j}=3e^{0}+\sum_{\alpha\in R}e^{\alpha}.$$

The only dominant root is $\varepsilon_1-\varepsilon_3=\omega_1+\omega_2$. Indeed, the simple roots are not dominant, since

$$\alpha_1=2\omega_1-\omega_2,\qquad \alpha_2=-\omega_1+2\omega_2,$$

and negative roots are not dominant. Thus $\omega_1+\omega_2$ and 0 are the only dominant weights of this tensor product. Example 17.8 gives their multiplicities 1 and 2 in $L(\omega_1+\omega_2)$:

dominant weight	$\omega_1+\omega_2$	0
$\operatorname{ch}\bigl(L(\omega_1)\otimes L(\omega_2)\bigr)$	1	3
$-\operatorname{ch} L(\omega_1+\omega_2)$	-1	-2
remainder	0	1
$-\operatorname{ch} L(0)$	0	-1
remainder	0	0

Hence

$$L(\omega_1)\otimes L(\omega_2)\cong L(\omega_1+\omega_2)\oplus L(0),\qquad 9=8+1.$$

This agrees with $\mathbb{C}^3\otimes(\mathbb{C}^3)^*\cong\mathfrak{sl}_3\oplus\mathbb{C}$ from Lecture 14.

Theorem 17.12 (Freudenthal's formula · *C*)**.** Humphreys, §22.3. *Let $\mathfrak{g}$ be simple and $\lambda\in P_+$. Let $(\ ,\ )$ be the form on $\mathfrak{h}^*$ induced by the Killing form, or any positive multiple of it. Then, for every $\mu\in P$,*

$$\bigl((\lambda+\rho,\lambda+\rho)-(\mu+\rho,\mu+\rho)\bigr)m_\lambda(\mu)=2\sum_{\alpha\in R_+}\sum_{k\geq 1}(\mu+k\alpha,\alpha)\,m_\lambda(\mu+k\alpha).$$

Humphreys states the formula with the Killing form. Both sides are linear in the form, so positive multiples are allowed. For a weight $\mu \neq \lambda$ of $L(\lambda)$, the left bracket is nonzero (loc. cit.). Once the weight set of $L(\lambda)$ is known, for example from Kostant's formula, the formula therefore gives a recurrence downward from λ. Its sums run over positive roots rather than over W, which helps when W is large. Nothing in this course depends on it.

As a check, take $\lambda = \omega_1 + \omega_2$ and $\mu = 0$, and scale the form so that all roots have squared length 2. Then $\lambda + \rho = 2\rho$, with $(\rho, \rho) = 2$ and $(2\rho, 2\rho) = 8$. Only $k = 1$ contributes, since 2α is not a weight. Each of the three positive roots then contributes $(\alpha, \alpha)\, m_\lambda(\alpha) = 2$. The formula becomes

$$(8 - 2)\, m_\lambda(0) = 2 \cdot 3 \cdot 2,$$

so $m_\lambda(0) = 2$ again.

Self-checks

(1) Why can $\dim L(\lambda)$ not be found by putting every $e^\mu = 1$ in $A_{\lambda+\rho}/A_\rho$?

(2) Compute $\dim L(2\omega_1 + \omega_2)$ and $\dim L(4\omega_1)$ for $\mathfrak{sl}_3$. What does the comparison show?

(3) Why is $m_\lambda(\mu) = 0$ in Kostant's formula when $\lambda - \mu \notin Q$?

Answers. (1) Both alternants map to $\sum_w \det(w) = 0$, so the evaluation gives $0 = 0$. A regular direction is needed. (2) $\frac{3 \cdot 2 \cdot 5}{2} = 15$ and $\frac{5 \cdot 1 \cdot 6}{2} = 15$. Nonisomorphic irreducibles can have equal dimension. (3) Each reflection changes a weight by an integer multiple of a root. Hence $w(\lambda + \rho) - (\mu + \rho) \equiv \lambda - \mu \pmod{Q}$. Since $\lambda - \mu \notin Q$, no argument of p lies in Q, so none lies in Q_+.

Summary

A ring homomorphism Φ_ρ to power series turns the character formula into a comparison of leading terms. Because ρ is regular, this gives the Weyl dimension formula, a product of ratios of coroot pairings over the positive roots.

Inverting the denominator in the completed character ring turns the character formula into Kostant's formula. It is an alternating sum of $|W|$ partition-function values, and the support of p discards most terms.

By complete reducibility, subtracting irreducible characters from the top dominant weight down decomposes a module. Lecture 18 compares these formulas with Schur functions in type A.

Reading

J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, §22.3 (Freudenthal's formula), §24.2 (Kostant's multiplicity formula) and §24.3 (Weyl's formulas). A. Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras* (2008), §§8.5–8.6. B. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 10. Exercise Sheet 17 contains the type C_2 calculations.